

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD3074 DEEP LEARNING FOR DATA SCIENCE

SATURDAY, 7 OCTOBER 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instructions to Candidates:

This question paper consists of FIVE (5) questions .

Answer **ALL** questions in **Section A** and **ONLY ONE(1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in section B, marks will only be awarded for the FIRST question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD3074 DEEP LEARNING FOR DATA SCIENCE**SECTION A (Answer all questions)**

Q1. (a) Identify the type of neural network that is suitable for each of the following tasks:

- (i) Count the number of people in an image. (2 marks)
- (ii) Classify if an English text is grammatically sound and correct. (2 marks)
- (iii) Predict the probability of graduate admission to a university based on an applicant's undergraduate result, GRE score, TOEFL score and the university ranking. (2 marks)

(b) Given the following function, find x that minimizes $f(x)$. Gradient Descent is used to minimize the function. The current value of $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

$$f(x_1, x_2) = x_1^2 + x_2^2 - x_1x_2$$

- (i) Draw the computational graph for $f(x_1, x_2)$. Use square (^), addition (+), minus (-) and multiply (*) as the basic computational units. (5 marks)
- (ii) Given the current value of x , perform forward propagation to compute the value of $\hat{y}$. (4 marks)
- (iii) Given the current value of x , perform backpropagation to compute the gradients $\frac{df}{dx_1}$ and $\frac{df}{dx_2}$. (Hints: $\frac{d}{dx} x^2 = 2x$, $\frac{d}{dx} xy = y$, $\frac{d}{dy} xy = x$, $\frac{d}{dx} x + y = 1$, $\frac{d}{dy} x + y = 1$, $\frac{d}{dx} x - y = 1$, $\frac{d}{dy} x - y = -1$) (7 marks)
- (iv) If we *increment* x_1 by Δx_1 , explain how it would affect the value of f . (3 marks)

[Total : 25 marks]

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- Q2. (a) (i) Specify the difference between *self-supervised* learning and the following learning paradigms:
- *supervised* learning
 - *unsupervised* learning (4 marks)
- (ii) Suggest TWO (2) pre-text tasks to train pre-trained NLP model using self-supervised learning. (4 marks)
- (iii) Suppose you are building a chat bot for your company's website. Why is it *not* recommended to train your model with a pre-trained model that was trained on classical Shakespeare poems? Notes: Shakespeare is an English poet in the 16th century. (2 marks)
- (b) The task is to detect the presence of defects in a water bottle manufacturing plant. We have trained with three different network architectures for 100 epochs with a batch size of 64 on the *same* training set. The following figure shows the accuracy of the models.

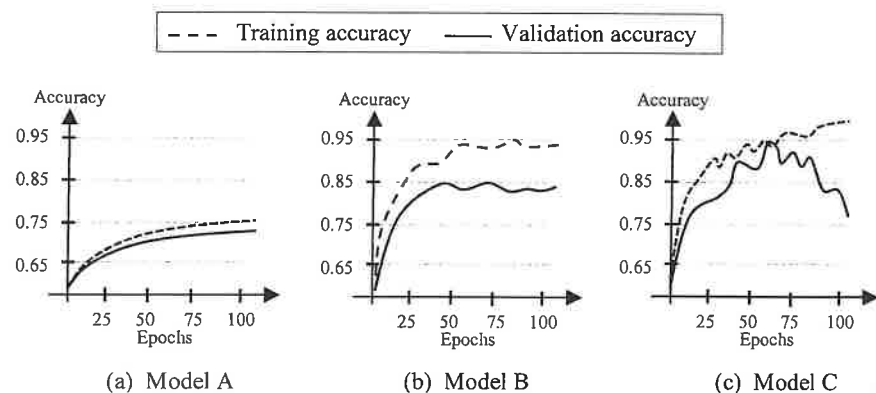


Figure 2.1: Training vs validation accuracy for Models A, B and C

- (i) Identify the issues with each Model A, B and C, respectively. (3 marks)
- (ii) Discuss if adding new training set may improve the validation accuracy of each of the models (Models A, B and C). (3 marks)
- (iii) Discuss if increasing the depth of the network may improve the performance of each of the models (Models A, B and C). (3 marks)
- (iv) Would you suggest reducing the batch size from 64 to 8 when training Model C? Justify your answer. (3 marks)
- (v) Based on the cost plots of Model A, B and C above, how would you select the final model for deployment? (3 marks)

[Total : 25 marks]

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- Q3. (a) Figure 3.1 shows the input activation map I and output activation map O of a single sample to a Conv2D layer. The kernel size $f=3$ and the batch size = 4.

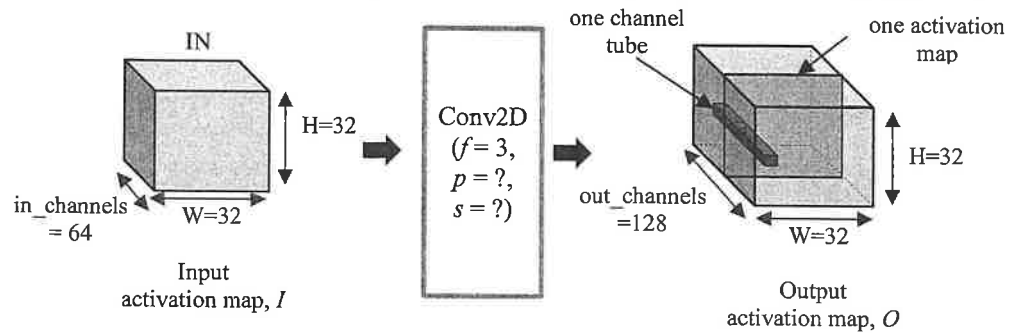


Figure 3.1: Convolutional Layer
(f : filter size, p : padding, s : stride)

- (i) How many filters are used in the layer? (3 marks)
 - (ii) Based on your answer in (a)(i), what is the shape of the tensor representing one filter? Use the Pytorch data format and identify the dimensions in your answer. (3 marks)
 - (iii) What are the settings for the stride s and padding p of the Conv2D layer in Figure 3.1? (3 marks)
 - (iv) Explain the difference between how one *output activation map* $O[c, :, :]$ and one *channel tube* $O[:, h, w]$ are generated by Conv2D. (4 marks)
 - (v) Suppose you replace the Conv2D in Figure 3.1 with a *pooling* layer with $f=4$, $s=2$ and $p=0$. What is the output volume shape? Use the Pytorch data format and identify the dimensions in your answer. (3 marks)
 - (vi) Suppose you replace the Conv2D in Figure 3.1 with *global pooling* layer. What is the output volume shape? Use the Pytorch data format and identify the dimensions in your answer. (3 marks)
- (b) Given the following 1-channel input volume to a Conv2D layer,

$$A_{\text{in}}$$

3	2	0	2
-2	1	4	1
-1	2	3	0
0	1	-2	-3

compute the output of *conv2D* layer ($s = 1$, $p = 0$) with the following kernel:

$$W$$

2	0
0	1

$$b$$

1

(6 marks)
[Total : 25 marks]

UCCD3074 DEEP LEARNING FOR DATA SCIENCE**SECTION B (Choose ONE question only)**

- Q4. The following diagram shows three **channels** representing the biosignals of a human indicating his/her health condition. The ECG records the electrical signal from the heart, the PPG detects volumetric changes in the blood circulation while the RIP measures the movement of the chest and abdominal wall.

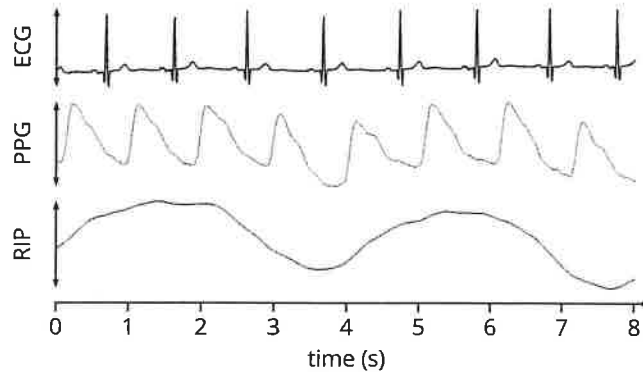


Figure 4.1: Biosignal input with three channels (ECG, PPG and RIP signals)

The task is to build an LSTM network to detect the presence of any *abnormal* biosignal pattern in the input sequence. The input is the biosignal input sequence while the output of the network is a binary value (1: abnormal, 0: normal).

- (a) Answer the following questions on the LSTM network.
- Justify why LSTM network can be used to model the task. (3 marks)
 - What is the suitable LSTM architecture for the task? Justify your answer. (3 marks)
 - Model A accepts inputs with all three channels (i.e., ECG, PPG and RIP). Model B accepts only one channel (e.g., ECG only). Compare the performance of the two models. (4 marks)
- (b) Suppose that each batch data contains the biosignals of 4 patients for training. For each patient, you collect the data for 1000 time steps. Suppose that a `batch_first=True` configuration is used.
- What is the shape of the batch data $X^{(t)}$? Use the Pytorch data format and identify the dimensions in your answer. (3 marks)
 - What is the shape of the output $\hat{Y}^{(t)}$? Use the Pytorch data format and identify the dimensions in your answer. (3 marks)
 - Design an LSTM neural network to model the task. Show your answer in the form of the block diagram of the unrolled version of the network. (9 marks)
- [Total : 25 marks]

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Q5. (a) The following figure shows the Encoder Network for a Transformer Model.

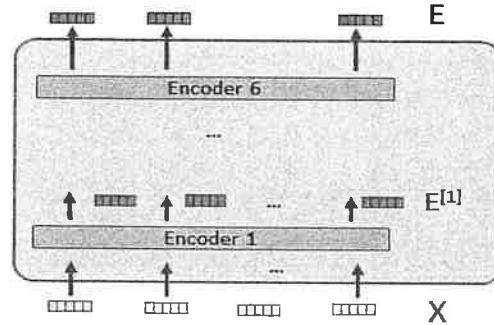


Figure 5.1: The Encoder Network for a Transformer Model

- (i) The layer is configured such that $d_model = 128$ and $batch_first = True$. For the input batch data, batch size $B = 4$ and the sequence length $T = 50$. Identify the correct shape for the following signals. Use the PyTorch data format and identify the dimensions in your answer:
- X : the input data X
 - $E^{[i]}$: the output of the encoder at layer i
 - E : the output of the encoder
- (3 marks)
- (ii) Identify if the following settings are fixed when defining the network, or can change during runtime.
- B : batch size
 - T : sequence length
 - d_model : data dimensional d_model
 - $batch_first$: batch first data format
- (4 marks)
- (iii) Explain why the Encoder Network stacks multiple layers of encoder block.
- (3 marks)
- (b) The following module shows the self-attention module in a Transformer layer. The layer is configured such that $d_model = 128$ and $batch_first = True$. For the input batch data, batch size $B = 4$ and sequence length $T = 50$.

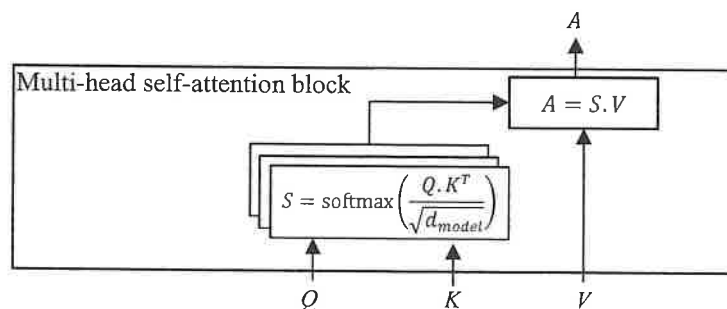


Figure 5.2: The multi-head self-attention block

UCCD3074 DEEP LEARNING FOR DATA SCIENCE**Q5. (b)(Continued)**

- (i) What is the role of the multi-head self-attention block? (3 marks)
- (ii) Why do we need multiple heads? (3 marks)
- (iii) Suppose that $S[t, :]$ is the row t in S . What do the values in $S[t, :]$ mean? (3 marks)
- (iv) What is the expected shape of the following signals? Use the PyTorch data format and identify the dimensions in your answer.
- S
 - Q, K and V (notes all have the same shape) (6 marks)
- [Total : 25 marks]
-

UCCD3074 DEEP LEARNING FOR DATA SCIENCE**Appendix: Formulas**Logistic Regression:

$$z = w_1 x_1 + \dots + w_{n_x} x_{n_x} + b \quad (\text{non-vectorized})$$

$$Z = w^T X + b \quad (\text{vectorized})$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (-y^{(i)} \ln \hat{y}^{(i)} - (1 - y^{(i)}) \ln(1 - \hat{y}^{(i)}))$$

$$w := w - \alpha \frac{\partial J(w, b)}{\partial w}$$

$$b := b - \alpha \frac{\partial J(w, b)}{\partial b}$$

Backpropagation for Logistic Regression:

(Non-vectorized)

$$dz^{(i)} = \hat{y}^{(i)} - y^{(i)}$$

activation phase

$$dw_j^{(i)} = x_j^{(i)} dz^{(i)}$$

summation phase

$$db^{(i)} = dz^{(i)}$$

summation phase

(Vectorized)

$$dZ = \hat{Y} - Y$$

activation phase

$$dw = \frac{1}{m} (X dZ^T)$$

summation phase

$$db = \text{np.mean}(dZ)$$

summation phase

Standard Neural Network:

$$Z^{[i]} = W^{[i]} A^{[i-1]} + b^{[i]}$$

summation phase for $i = 1 \dots L$

$$A^{[i]} = g(Z^{[i]})$$

activation phase for $i = 1 \dots L$

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

softmax function

$$J(\hat{Y}, Y) = \frac{1}{m} \sum_{i=1}^m -\ln(\hat{y}_{y^{(i)}}^{(i)})$$

cross entropy function where $\hat{y}_{y^{(i)}}^{(i)}$ is the probit score $\hat{y}^{(i)}$ for the true class $y^{(i)}$

(Activation function)

$$g(z) = \frac{1}{1+e^{-z}}$$

sigmoid activation

$$g'(z) = g(z)(1 - g(z))$$

$$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

tanh activation

$$g'(z) = 1 - g(z)^2$$

$$g(z) = \max(0, z)$$

relu activation

$$g'(z) = \begin{cases} 1 & \text{if } x > 0 \\ \text{undefined} & \text{if } x = 0 \\ 0 & \text{if } x < 0 \end{cases}$$

$$g(z) = \max(0.01z, z)$$

leaky relu activation

$$gg'(z) = \begin{cases} 1 & \text{if } x > 0 \\ \text{undefined} & \text{if } x = 0 \\ 0.01 & \text{if } x < 0 \end{cases}$$

Backpropagation for the Standard Neural Network:

$$dZ^{[i]} = dA^{[i]} * g'^{(i)}(Z^{[i]})$$

Activation phase

$$g^{[i]'}(Z^{[i]}) = A^{[i]} * (1 - A^{[i]})$$

Activation phase (for sigmoid activation)

$$g^{[i]'}(Z^{[i]}) = Z^{[i]} > 0$$

Activation phase (for relu activation)

$$dW^{[i]} = \frac{1}{m} dZ^{[i]} A^{[i-1]T}$$

Summation phase

$$db^{[i]} = \text{np.mean}(dZ^{[i]})$$

Summation phase

$$dA^{[i-1]} = W^{[i]T} dZ^{[i]}$$

Summation phase

Convolutional Neural Network:

$$O(i, j) = W \cdot X_{loc}(i^*, j^*) + b$$

Activation value at location (i,j)

$$\text{size} = \left\lfloor \frac{i+2p-f}{s} + 1 \right\rfloor$$

size of the activation map

$$\# \text{params in a filter} = c_f \times h_f \times w_f + 1$$

Number of parameters in a filter

$$\# \text{params in a conv layer} = \# \text{filters} \times \# \text{params in a filter}$$

Number of parameters in a CONV layer

$$\# \text{computations} = \# \text{activations in output} \times \# \text{params in a filter}$$

Number of computations in a CONV layer

$$= \text{size of output activation map} \times \# \text{params in a filter}$$

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depth $d = \alpha^\phi$, width $w = \beta^\phi$, resolution $r = \gamma^\phi$
 subject to: $\alpha, \beta^2, \gamma^2 \approx 2$
 $\alpha \geq 1, \beta \geq 1, \gamma \geq 1$

Efficient Net formulation

RNN:

$$h^{<t>} = g_1(W_{hh}h^{<t-1>} + W_{hx}x^{<t>} + b)$$

$$h^{<t>} = g_1\left(W_h\begin{bmatrix} h^{<t-1>} \\ x^{<t>} \end{bmatrix} + b\right)$$

Simple RNN Cell

Simple RNN Cell (Simplified form)

LSTM:

$$f^{<t>} = \sigma(W_f(h^{<t-1>}, x^{<t>})) + b_f$$

$$i^{<t>} = \sigma(W_i(h^{<t-1>}, x^{<t>})) + b_i$$

$$o^{<t>} = \sigma(W_o(h^{<t-1>}, x^{<t>})) + b_o$$

$$\tilde{c}^{<t>} = \tanh(W_c(h^{<t-1>}, x^{<t>})) + b_c$$

$$c^{<t>} = i^{<t>} * \tilde{c}^{<t>} + f^{<t>} * c^{<t-1>}$$

$$h^{<t>} = o^{<t>} * \tanh(c^{<t>})$$

forget gate

input gate

output gate

new content (raw)

cell state

hidden state

Attention-based RNN:

$$h_d^{<t_d>} = g(W_{hh}h^{<t_d-1>} + W_{hx}x^{<t_d>} + W_{hc}c^{<t_d>} + b)$$

$$e^{<t_d, t_e>} = \text{fc}(h_e^{<t_e>}, h_d^{<t_d-1>}),$$

$$\alpha^{<t_d, t_e>} = \frac{\exp(e^{<t_d, t_e>})}{\sum_{t_e=1}^T \exp(e^{<t_d, t_e>})},$$

$$c^{<t_d>} = \sum_{t_e=1}^T \alpha^{<t_d, t_e>} h^{<t_e>}$$

output hidden state of a decoder RNN cell

formula to generate the context vector $c^{<t_d>}$ in the attention module of the decoder at time t_d

Transformer:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_{\text{model}}}}\right)V$$

self-attention module (without masking)

$$A_{\text{masked}} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_{\text{model}}}} + M\right)V$$

self-attention module (with mask)

$$\tilde{X} = \frac{X - \text{mean}_{c,T}}{\text{std}_{c,T}}$$

Layer normalization for input X of shape (B, T, C), B: batch size, T: sequence length, C: channel

Training NN:

$$x_{\text{std}} = \frac{x - \mu}{\sigma}$$

Standardization

$$\hat{z}_c^{[t]} = \frac{z_c^{[t]} - \mu_c^{[t]}}{\sqrt{v_c^{[t]} + \epsilon}}, \quad \tilde{z}_c^{[t]} = \gamma_c \cdot \hat{z}_c^{[t]} + \beta_c$$

Batch normalization (BN) during training for batch t and channel c .

$$\mu_c^{[t]} = \frac{1}{BHW} \sum_{i=1}^B \sum_{h=1}^H \sum_{w=1}^W z_{c,h,w}^{[t]}$$

Mean of BN during training for an input of shape (B, H, W, C)

B: batch size, H: height, W: height, C: channel size

Variance of BN during training for an input of shape (B, H, W, C)

B: batch size, H: height, W: height, C: channel size

Running mean for BN during training

$$\vec{\mu}_c = (1 - m)\vec{\mu}_c + m\mu_c^{[t]}$$

$$\vec{v}_c = (1 - m)\vec{v}_c + m v_c^{[t]} \frac{BHW}{BHW - 1}$$

Running variance for BN during training

$$J(W^{[1]}, b^{[1]}, \dots, W^{[L]}, b^{[L]}) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}, y) + \lambda \frac{1}{2m} \sum_{l=1}^L \|W^{[l]}\|_2^2$$

L2 regularization for an L -layered network, m samples

$$J(W^{[1]}, b^{[1]}, \dots, W^{[L]}, b^{[L]}) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}, y) + \lambda \frac{1}{2m} \sum_{l=1}^L |W^{[l]}|$$

L1 regularization for an L -layered network, m samples

$$\text{var}(W^{[l]}) = \frac{2}{(n^{[l-1]} + n^{[l]})}$$

Xavier initialization

$$\text{var}(W^{[l]}) = \frac{2}{n^{[l-1]}}$$

Kaiming initialization

